

Balance, Not Battle: Rethinking Acne Through the Microbiome

Dermatology AI Interest Group, Monthly Spotlight: Acne

What Acne Actually Is ?

Acne or as it's known scientifically, "acne vulgaris," is far more than an occasional pimple that irritates many and may be a nightmare for others. It is a chronic inflammatory condition of the pilosebaceous unit, which is the hair follicle and its attached oil gland, and it's considered one of the most common inflammatory skin conditions affecting patients among all age groups, with different underlying causes and ways of treatment, making the understanding of this whole condition much more complicated for individuals with limited access to the actual whole story. Understanding the actual scientific meaning and underlying causes of acne might make it easier to control and treat it while minimizing both the financial burden and the psychosocial impact on the patient.

For decades, textbooks rested on 4 main pillars, each of which leads to the other:

- . Excess sebum production.
- Abnormal skin shedding leading to hair follicle blockage
- Colonization of the main acne causative bacteria, the *Cutibacterium acnes*.
- The inflammatory and immune reaction followed by this cycle.

Now you might have thought about hormones ruling in this scenario, and let us break it to you that hormones, whether they were produced during puberty or due to metabolic imbalance, work their way through, leading to acne, by contributing to this 4 pathway cycle through their effect on IGF-1 (insulin-like growth factor-1) and androgen overproduction causing an overdrive effect on sebum production and blocking of its protective regulatory protein called FoxO1; the blockage of FoxO1 exacerbates the sebum production even more, leading to the 4 mentioned pattern: Oil overproduction → Inflammatory signaling switched ON → Skin tries to compensate by increasing its turnover activity around the hair follicle, where the sebaceous gland is located → Resulting in pathogenic bacterial colonization.

The severity of this cycle, when it's caused by hormonal imbalance, relies hugely on each individual's genetic profile, which explains why patients differ in their response to the same factors leading to acne.

The Story Behind the Microbiome's Role in Acne.

With the increasing reports of antibiotic resistance reached by patients using antibiotics to treat acne, the modern picture of investigating the actual role of the acne-related bacteria became a must to control this issue. What was found is that acne tracks much more closely

with a loss of microbial diversity, a state generally called dysbiosis, than with *C. acnes* simply being abundant.

C. acnes itself turns out to be a diverse species, with certain strains increasing under specific conditions that drive acne. Due to its diversity, one lineage in particular, referred to as IA1, stood out in multiple studies as being associated with acne lesions. The acne-associated lineages tend to have a whole aggressive genetic toolkit for acne causation: - Toxins with the capability of producing holes in skin cell membranes - Pigments that can generate ROS (reactive oxygen species), which can damage skin tissues - Enzymes that can break down the skin's structural matrix - Biofilm production, which interferes with treatment methods (Biofilms are bacterial colonies that stick tightly to hair follicles, preventing antibiotic treatment from penetrating and reaching its target)

Notably, the non-acne-associated lineage of *C. acnes* tends to be found in healthy skin, living peacefully without any aggressive toolkit.

In the microbiome community, other bacteria, such as *Staphylococcus epidermidis*, are also considered among the most common and least trouble-causing residents, playing a quiet but major role in controlling other residents in this community, such as *C. acnes*.

Under balanced skin status, *S. epidermidis* tends to reduce the inflammation and damage that's provoked by *C. acnes*, partly through metabolic byproducts like succinic acid. Moreover, other *Staphylococcus* species, such as *S. capitis*, have been shown experimentally to suppress *C. acnes* more effectively than some conventional antibiotics, without harming the rest of the skin's bacterial community. On the other hand, other microorganisms, such as the lipophilic fungus *Malassezia*, which thrives in oily, sebum-rich skin, have also been linked to flare-ups in some acne patients, and using anti-fungal treatments has been shown to provide clinical improvements in some cases.

Taken together, these findings help us reframe the actual microbial causative agent scientifically as a cooperative microbial community, one in which the loss of diversity and the rise of specific aggressive lineages disturb a balance that healthy skin normally maintains on its own. And by working on reaching the balance in skin microbiota, the new science of acne treatment has entered the field.

Reaching Beyond the Skin: The Gut-Skin Axis

Imagine this typical and very common scenario, and understand the science behind it... You went through a vacation, had many sugary and fatty foods, you enjoyed yourself a bit too much, and now you wake up back home with a face filled with inflamed pimples all around, so you panicked and blamed it all on your messed-up diet, but have you ever thought why? How can food be the reason behind this flare-up?

A diet that is poor in nutrients and fiber affects the gut microbiota diversity, increasing bad strains and reducing the good ones. The pathogenic gut bacteria weaken the gut integrity, leading to leakage of their byproducts, such as lipopolysaccharide, which directly reaches the bloodstream, provoking low-grade systemic inflammation that doesn't stay trapped in the gut but reaches the skin, leading to multiple inflammatory skin conditions, acne -

included. In addition, for acne specifically, the nutrient-sensing cellular pathway that governs sebaceous gland activity, the mTOR pathway, also appears to be affected directly by the circulating gut bacterial byproducts and diet patterns that are high in glycemic load. These findings provide a better understanding of the long-term effect of our diet on skin health, and open up an even more interesting way of treatment: probiotics.

Let's talk treatment wisely:

Oral antibiotics such as doxycycline, minocycline, and lymecycline remain a mainstay of moderate-to-severe acne treatment, and they generally improve symptoms by reducing *C. acnes* numbers. Studies comparing the before and after skin state when using antibiotics for treatment show a reduction in *C. acnes*, accompanied by an increase in *S. epidermidis*, as well as restoration of skin microbiota diversity. These incidental findings, which aren't the main intent of using antibiotics in treating acne, may encourage patients to rely on them as their first choice of treatment. That is, until antibiotic resistance enters the conversation: with the increase in the use of antibiotics, *C. acnes* compensates by forming its own biofilm that can't be penetrated by antibiotic drugs. Resistance traits can be transferred horizontally to other, more dangerous organisms such as *Staphylococcus aureus*. On the other hand, topical antibiotic treatments tend to have a local effect on *C. acnes* without affecting skin microbiome diversity.

For more severe cases, isotretinoin is used, which has its effect directly on sebaceous gland sebum production without having any role in skin bacterial composition, yet it still significantly reduces *C. acnes* abundance as a downstream consequence, and detailed metabolic studies have found it alters bacterial amino acid and vitamin synthesis pathways on the skin.

Toward Microbiome-Sparing Treatment Probiotic-based therapies are the furthest along this path. *Lactobacillus plantarum*-GMNL6 and *Enterococcus faecalis* have shown the ability to suppress *C. acnes* proliferation and biofilm formation, without affecting the protective skin bacteria. Topical probiotic creams built around *Lactobacillus plantarum* were used and showed effectiveness in reducing bacterial proliferation and inflammation at treated sites. Moreover, bacteriophages, viruses that have been engineered to target *C. acnes* selectively, are being explored, though clinical evidence in humans is still very limited. Newer non-antibiotic drugs such as trifarotene and clascoterone have been approved specifically because they manage acne through sebum and inflammation control rather than direct antibacterial activity. Targeting diet patterns and skin care products is also considered part of the acne treatment plan, to ensure none of them are affecting the skin microbiota diversity negatively.

At the end of the day, acne isn't simply a war against bacteria; it's a battle for balance. The goal going forward isn't to wipe out *C. acnes* altogether, but to restore the microbial harmony that healthy skin already knows how to maintain on its own, through smarter antibiotic use, targeted probiotics, and skincare choices that work with our microbiome instead of against it.

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